

VastWaste

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1 PROJECT URL

<https://www.ozgesamanci.com/#/vastwaste/>



2 DESCRIPTION OF ARTWORK

Humans once perceived oceans as boundless, and thus impossible to pollute—until we created the Great Pacific Garbage Patch. The same pattern is now repeating in outer space. VastWaste is a data-driven, projection art installation that illuminates the parallels and interplay between marine pollution and space debris. It can also be experienced in Virtual Reality.

In VastWaste, marine pollution data generates the movements and collisions of space debris. The installation portrays two vast

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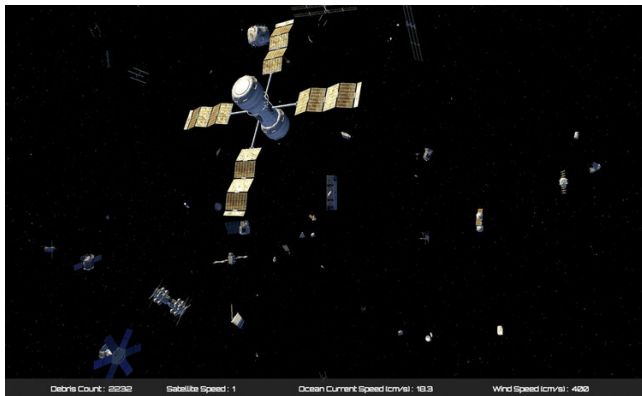
worlds: outer space and ocean depths. In the final scene, the byproducts of space exploration end up in the oceans. The destiny of these vast spaces merge in reality and in this data-driven experience.

Human activities have scattered millions of objects into Earth's orbit. With no friction, debris travel at 18,000mph. Even tiny paint flecks can create explosive crashes. Approximately 4,000 operational satellites are currently in Earth's orbit and the amount of space debris is already at a critical point. US and European Space Agencies track space debris and maneuver spacecraft to avoid collisions. SpaceX's Starlink plans to add 40,000 satellites in the next decade, yet there is no known solution for mitigating the space debris.

If the amount of space debris passes a critical mass, each collision will lead to more collisions in a chain reaction, known as the Kessler Effect. Ultimately, future spacecraft launches from Earth may become impossible. VastWaste generates an ever-changing Kessler Effect in conjunction with a data-driven soundtrack.

Satellites spin based on the speed of marine debris. This is calculated using ocean currents and ocean winds. The number of fragments falling into the ocean is tied to human use of satellites, symbolized by the number of tweets per second. Generative music varies in each play based on collisions, number of fragments, and their descent into the ocean.

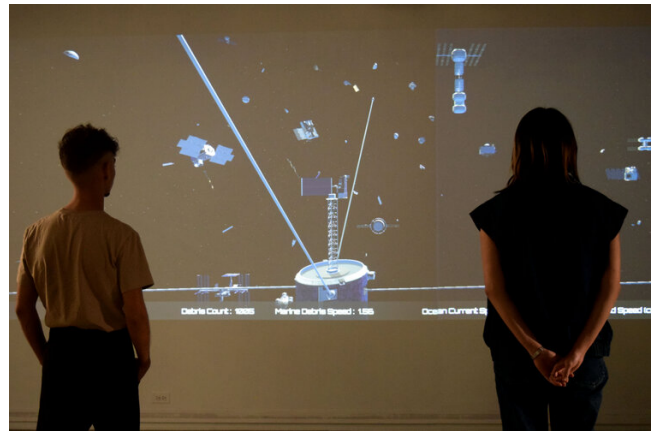
Humans observe marine pollution with satellites, and we bury dead satellites into our oceans. The future of two vast spaces is entangled.



3 ARTIST INFORMATION

Ozge Samanci, media artist and graphic novelist, is an associate professor in Northwestern University's School of Communication. Her interactive installations have been exhibited internationally, including Siggraph Art Gallery, FILE festival, Currents New Media, The Tech Museum of Innovation, WRO Media Art Biennial, Athens International Festival of Digital Arts and New Media, Píksel Electronic Arts Festival, ISEA among others.

Jenni Hutson is a software engineer interested in exploring alternative modes of interaction with the environment. She received a degree in film and computer science from Northwestern University and will begin graduate studies in Media Arts and Technology at UC Santa Barbara this fall. She is a computer programming lead for VastWaste.



Jonathan David is a graduate of Northwestern University and a software engineer. He is a programmer and implementer for the VastWaste project.

Wren Gotschalk is studying computer science at Northwestern University. Wren is a programmer and implementer for the VastWaste project.

Jack Burkhardt is a Northwestern University student whose interests lie broadly in human-computer interaction and interactive entertainment. He is a programmer and implementer for the VastWaste project.

Rachel Kantor is a software engineer and environmental scientist. She received a degree in computer science from Northwestern University and is a data researcher for VastWaste.

Stephan Moore is a sound artist, designer, composer, improviser, programmer, engineer, teacher, and curator based in Chicago. His creative work manifests as electronic studio compositions, improvisational outbursts, sound installations, scores for collaborative performances, algorithmic compositions, interactive art, and sound designs for unusual circumstances.

Liza Salvi is a game developer who thrives at the intersection of code, art, and the surreal. She studied Computer Science and Media Arts at Northwestern University, and currently works as a software engineer and gameplay programmer for Lost Boys Interactive.